Google Play Store App Rating Prediction

Project Overview

This project builds a linear regression model to predict Google Play Store app ratings using app metadata. The goal is to help Google identify promising apps for boosted visibility in recommendations and search results.

Dataset

- **Source:** Google Play Store dataset (googleplaystore.csv)
- **Original size:** 10841 records, 13 features
- **Final size after cleaning:** 8496 records

Steps Performed

1. Data Loading & Cleaning: Null values were dropped, and key columns were reformatted: Size (MB/KB converted to numeric), Reviews (string to float), Installs (symbols removed, converted to integer), and Price ($ symbol removed, converted to float). "Varies with device" entries in the Size column were replaced with the median to preserve data integrity.

2. Sanity Checks: Validated that ratings fell between 1–5, reviews did not exceed installs, and free apps had a price of zero.

3. Univariate Analysis: Boxplots and histograms revealed that most apps are free, lightly reviewed, small, and rated between 4.0 and 5.0.

4. Outlier Treatment: Apps priced above $200, apps with more than 2 million reviews, and apps exceeding the 95th percentile of installs (10 million) were removed.

5. Bivariate Analysis: Scatter plots and boxplots revealed no strong relationship between rating and price, size, or reviews. The Events and Health & Fitness categories showed the highest median ratings.

6. Data Preprocessing: Log transformation was applied to Reviews and Installs to reduce skewness. Irrelevant columns were dropped and categorical variables were one-hot encoded, resulting in 157 features.

7. Model Building: A linear regression model was trained on a 70/30 train-test split.

Results

Metric Score

R^2 Train Set	0.107
R^2 Test Set	0.087

Conclusion

The model explains approximately 10% of the variance in app ratings, indicating weak but consistent predictive power. The low R^2 suggests that app ratings are influenced by subjective factors not captured in the available features. Future improvements could include using non-linear models or incorporating additional features.